

Mathematics, 2013 EC, entrance exam, version 1

Grade 12 · Mathematics · 2013 E.C.

Time allowed: 120 minutes

Questions: 36

Total marks: 36

Name _____ Grade _____ ID No. _____ Date _____

1. Let a function $f(x) = -4x^3 - 2x - 10$, then which one of the following is true about f . (1)
 - A. f is neither even nor odd
 - B. f is odd
 - C. f is even
 - D. f is both even and odd
2. If $(g \circ f)(x) = x^2 + x + 6$ and $f(x) = x + 2$, then what is the equivalent expression for $g(x)$? (1)
 - A. $x^2 - 3x + 8$
 - B. $-x^2 - 3x + 6$
 - C. $x^2 + 3x + 8$
 - D. $x^2 - 3x - 6$
3. Which one of the following is incorrect about the relation $R = \{(x,y): y = (x+3)/(x+10)\}$? (1)
 - A. The range of R is $(-\infty, 1) \cup (1, \infty)$
 - B. Domain of R is $(-\infty, -10) \cup (-10, \infty)$
 - C. $R = \{(x,y): y = (x+3)/(x+10)\}$ can be a function
 - D. The inverse of R is $\{(y,x): y = (10x+3)/(x+1)\}$
4. Which one of the following is correct one-to-one correspondence? (1)
 - A. $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = \log_a x, \forall x \in \mathbb{R}$
 - B. $f: \mathbb{R} \rightarrow \mathbb{Q}$ given by $f(x) = 4^x, \forall x \in \mathbb{R}$
 - C. $f: [0, \infty) \rightarrow [0, \infty)$ given by $f(x) = x^2, \forall x \in \mathbb{R}$
 - D. $f: [1, \infty) \rightarrow (3, \infty)$ given by $f(x) = -3 + \sqrt{x-1}, \forall x \in \mathbb{R}$
5. Which one of the following is not a rational expression? (1)
 - A. $(x^5+9)/(x^2-16)$
 - B. $\sqrt{((x+2)^2)/(x^2-4)}$
 - C. $\sqrt[3]{((x+2)^6)/(x^2-25)}$
 - D. $\sqrt[3]{((x+2)^{12})/(x^2+16)}$
6. What is the solution set of the expression $(36x-72)/(-x^3+4x) = [(x-1)/(x+1) - (x+1)/(x-1)] / [(x+2)/(x^2-1)]$? (1)
 - A. $\{-3, -2\}$
 - B. $\{-3, 3\}$
 - C. $\{3, 2\}$
 - D. $\{3, 3\}$
7. Which one of the following is incorrect about the parabola $y^2 + 16x - 6y - 23 = 0$? (1)
 - A. its directrix is $x=6$
 - B. its axis of symmetry is $y=3$
 - C. its vertex is $(3, 2)$
 - D. its focus is $(-2, 3)$
8. Let $(-6, -4)$ & $(2, 2)$ are end points of the diameter. Which one is the equation of the circle? (1)
 - A. $x^2 + y^2 + 4x + 2y - 30 = 0$
 - B. $x^2 + y^2 + 4x + 2y + 20 = 0$
 - C. $x^2 + y^2 + 4x + 2y = 20 = 0$
 - D. $x^2 + y^2 + 4x + 2y = 30 = 0$
9. What are the intersection points of the line and the circle, with equation $(x-1)^2 + (y+1)^2 = 100$ and the line $8x - 6y = 14$? (1)
 - A. $(8, 9)$ and $(-8, -8)$
 - B. $(8, 9)$ and $(8, 8)$
 - C. $(7, 7)$ and $(-5, -9)$
 - D. $(-7, 7)$ and $(5, -9)$

10. Which one of the following is false about the ellipse $25x^2+16y^2+200x+64y+64=0$? (1)

- A. its center is $C(-4,-2)$
B. its major axis is vertical
C. its foci are $(-4,-5)$ & $(-4,1)$
D. its vertices are $(-4,-5)$ & $(-5,-2)$

11. Which one of the following sentences is not a proposition? (1)

- A. Plant produces their own food
B. Welcome!
C. There is no medicine for HIV/AIDS.
D. Man is mortal

12. If $(p \Leftrightarrow \neg q) \Rightarrow (\neg p \vee r)$ is False, then which of the following has a truth value false? (1)

- A. $\neg q \Leftrightarrow r$
B. $q \Rightarrow p$
C. $q \vee \neg r$
D. $\neg p \Leftrightarrow r$

13. Which one of the following pairs of compound propositions are not equivalent propositions? (1)

- A. $p \Rightarrow q \equiv \neg p \vee q$
 B. $\neg(p \vee q) \equiv \neg p \wedge \neg q$
 C. $\neg(p \vee q) \equiv \neg p \wedge \neg q$
 D. $p \Rightarrow q \equiv \neg q \Rightarrow \neg p$

14. Then which one of the following quantifier has a truth value false, $\forall x, y \in \mathbb{R}$? (1)

- A. $(\forall x)(\forall y)(x^2+y^2 \geq 0)$ B. $(\exists x)(\exists y)(x^2+y^2 < 0)$
C. $(\exists x)(\exists y)(x^2+y^2 = 0)$ D. $(\forall x)(\forall y)(-x^2-y^2 \leq 0)$

15. A committee of 8 students has to be formed from 6 boys and 4 girls. What is the probability that the committee contains 5 boys and 3 girls? (1)

- A. $\frac{2}{13}$
B. $\frac{5}{7}$
C. $\frac{8}{15}$
D. $\frac{2}{3}$

16. A bag contains 4 red, 5 yellow and 3 green balls. Three balls are drawn without replacement. Probability of red first, yellow second, green third? (1)

- A. 12/22 B. 1/22
C. 7/22 D. 9/22

17. What is the constant term in the expansion of $(-2x + 5/x^5)^6$? (1)

- A. 160
B. -960
C. -160
D. -960

18. How many three-digit numbers can be formed from the digits 2,3,5,7,9 if each digit is used once only? (1)

- A. 60
B. 125
C. 100
D. 50

19. If $A = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 2 & -1 \\ -2 & 3 & 1 \end{bmatrix}$. Then what is the determinant of $2A$? (1)

- A. 32
C. 64
- B. -32
D. -64

20. What is the value of x , if $|[2+x, 0, 0], [0, x+2, 0], [0, 0, x+2]| = 125$? (1)

- A. -3
B. 3
C. 5
D. -5

21. Let system: $x+2y+z=2$, $-x+y-2z=1$, $bx+y+z=1$. Which one is not true? (1)
- A. If $B \neq 1/5$, then the system has no solution
 B. If $B \neq 4/5$, then the system has unique solution
 C. If $B=4/5$, then the system has infinitely many solution
 D. If $B=1/5$, then the system has unique solution
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22. Let $Z = (2-3i)^4 / [(3+4i)(5-12i)]$. Then what is the magnitude of Z ? (1)
- A. $169/\sqrt{65}$ B. $169/65$
 C. $13/5$ D. $13/\sqrt{65}$
-
23. Let $Z = ((\sqrt{6}+\sqrt{6}i)/(1+\sqrt{3}i))^{15}$. Which one is true about Z ? (1)
- A. $\text{Arg } z = 3\pi/4$
 B. The modulus of Z is 3^{15}
 C. The polar form of Z is $3^{15/2}(\cos(-15\pi/12)+i \sin(-15\pi/12))$
 D. $\text{Arg } z = 15 \text{ Arg}((\sqrt{6}+\sqrt{6}i)/(1+\sqrt{3}i))$
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24. Let $Z = 2\sqrt{2} (\cos(11\pi/4) + i \sin(11\pi/4))$. What is the Cartesian representation? (1)
- A. $-2-2i$ B. $-2+2i$
 C. $2+2i$ D. $2-2i$
-
25. Let an equation of a circle is $(x-3)^2+(y-4)^2=64$. Then what is the equation of the image of circle after a reflection about the line $y = -x+5$? (1)
- A. $x^2+(y-5)^2=64$ B. $(x-1)^2+(y-2)^2=64$
 C. $(x-5)^2+y^2=64$ D. $(x-2)^2+(y-3)^2=64$
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26. What is the general solution of $2\cos^2 x + 3 \sin x = 0$? (1)
- A. $\{7\pi/6+2n\pi, 11\pi/6+2n\pi\}$ B. $\{-7\pi/6+2n\pi, 11\pi/6+2n\pi\}$
 C. $\{-7\pi/6+2n\pi, -11\pi/6+2n\pi\}$ D. $\{7\pi/6+2n\pi, -11\pi/6+2n\pi\}$
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27. Angle of elevation to top of building 60° , to a point 14m below top is 45° . Find height of building. (1)
- A. 30m B. 34m
 C. 24m D. 20m
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28. $\lim_{x \rightarrow 1} \{ \sqrt{(1+\sqrt{8x+1})} - 2 \} / (x-1) = ?$ (1)
- A. $-1/24$ B. $1/24$
 C. $1/3$ D. $-1/3$
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29. Let $f(x)=e^{4x+3}$ and $g(x)=\sin x$. If $F(x)=x f(x)+ f(g(x))$, then $F'(0)=?$ (1)
- A. $5/e^3$ B. $5e^3$
 C. $-5e^3$ D. $-5/e^3$
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30. If the region bounded by $y=2.5x^2$ and $y=10$ rotates about x-axis, volume? (1)
- A. 80π B. 320π
 C. 160π D. 300π
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31. Radius of cylinder height 10 cm increases at 6 cm/min. Rate of change of volume when radius=5 cm? (1)
- A. 200π B. 800π
 C. 500π D. 600π

32. Let $f(x) = (x^2 - 25)/|x - 5| = \{ x + 5 \text{ if } x > 5, \text{ undefined? if } x = 5, -x - 5 \text{ if } x < 5 \}$. Which is true? (1)

- A. f is continuous on $(-\infty, 5) \cup (5, \infty)$ B. f is continuous at $x = 5$
C. The limit of f exists D. f is continuous on $(-\infty, \infty)$
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33. $\lim_{x \rightarrow \infty} ((2x+5)/(2x-11))^{2x+1} = ?$ (1)

- A. e^{16} B. e^8
C. e^{-16} D. e^{-8}
-

34. Let $f(x) = e^x \cos x$. Then $f''(x) = ?$ (1)

- A. $-2e^x \cos x$ B. $2e^x \sin x$
C. $-2e^x \sin x$ D. $2e^x \cos x$
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35. Let $f(x) = \sqrt{64 - x^2}$. Then the largest interval that f is differentiable? (1)

- A. $(-8, 8)$ B. $[-8, 8]$
C. $(-\infty, -8] \cup [8, \infty)$ D. $(-\infty, -8) \cup (8, \infty)$
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36. Solve system: $x - 2y + z = 7$, $3x + y - z = 2$, $2x + 3y + 2z = 7$. Solution set? (1)

- A. $\{-2, -1, -3\}$ B. $\{2, -1, 3\}$
C. $\{-2, 1, -3\}$ D. $\{-2, -1, 3\}$
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— END OF EXAMINATION —